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i **ASSIST does not take the place of a counselor on your campus.** It is intended to help students and counselors work together to establish an appropriate path toward transferring from a public California community college to a California university.

Major Articulation Agreement

Chemical Engineering, B.S.

Effective during the **2025-2026** academic year

	To: University of California, Berkeley 2025-2026 General Catalog, Semester	←	From: De Anza College 2025-2026 General Catalog, Quarter
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COLLEGE ADMISSION REQUIREMENTS

The major in Chemical Engineering is offered by the College of Chemistry.

It is designed to equip the student for professional work in development, design, and operation of chemical products and processes. It prepares the student for employment in such industries as chemical, petroleum, electrochemical, biochemical, semiconductor, nuclear, aerospace, plastics, food processing, environmental control, or related industries.

Transfer applicants are expected to complete, at a minimum, coursework equivalent to Berkeley's:

CHEMISTRY 1A + 1AL + 1B
MATH 51 + 52 + 53 + 54
PHYSICS 7A + 7B
READING & COMPOSITION A

Coursework must be completed by the end of the spring term preceding fall enrollment at Berkeley.

IGETC/Cal-GETC is not required. Students who choose to complete the entire IGETC/Cal-GETC pattern by the end of the spring term preceding fall enrollment at Berkeley may use IGETC/Cal-GETC to fulfill the Reading and Composition Requirement.

Lower division courses required for graduation (but not for admission) are also listed in this articulation agreement. Completion of those courses is strongly recommended in order to strengthen one's application. All major courses must be taken for a letter grade. High grades in major courses (B and A grades exclusively) are essential for applicants to be both competitive in the admissions process and to be adequately prepared to continue with junior year coursework at Berkeley.

The applicant's Personal Insight Questions (PIQs) are important in the admissions process. The PIQs are reviewed for evidence of the student's interest in the chosen field and a thoughtful match between the intended major and academic and career objectives.

For more information on College of Chemistry policies and degree programs:

<https://chemistry.berkeley.edu/ugrad/degrees>

For more information on admission to UC Berkeley:

<https://admissions.berkeley.edu>

For more information on majors at UC Berkeley:

<https://undergraduate.catalog.berkeley.edu/>

Additional questions about transferring to the College of Chemistry may be addressed to:

Jahzeel Flores Morales, Director of Undergraduate Student Services
jahzeel@berkeley.edu
(510) 643-0550

ADDITIONAL REQUIREMENTS

BIOLOGY

Biology 1A (lecture only) is required for the Chemical Engineering B.S. degree.

You may also satisfy this requirement with a score/grade of:
4 or higher on the AP Biology exam satisfies BIOLOGY 1A & 1AL.

MATHEMATICS

You may also satisfy this requirement with a score/grade of:

3 or higher on the AP Calculus AB exam satisfies MATH 51;
3 or 4 on the AP Calculus BC exam satisfies MATH 51;
5 on the AP Calculus BC exam satisfies MATH 51 + 52.

ORGANIC CHEMISTRY WITH LABORATORY

CHEM 12A (organic chemistry) is required for the Chemical Engineering B.S. degree. CHEM 12B is not required for the Chemical Engineering major, but it is acceptable in satisfaction of the science elective for the open elective program.

Transfer students are encouraged to take the organic chemistry sequence at their institution.

Completion of Chem 3A + 3AL + 3B + 3BL, with a mean grade of B or better, will constitute satisfactory completion of Berkeley's Chem 12A + 12B. Students not meeting this requirement may be required to take Chem 12A and/or Chem 12B at Berkeley.

PHYSICS 7C is not required for the Chemical Engineering major, but it is acceptable in satisfaction of the science elective for the open elective program.

READING AND COMPOSITION REQUIREMENT

Coursework equivalent to Berkeley's: Reading & Composition A; or entire IGETC/Cal-GETC pattern completed by the end of the spring term preceding fall enrollment at Berkeley.

NOTE: You may also satisfy this requirement with a score/grade of: 4 or 5 on the AP exam in either English Language and Composition or English Literature and Composition.

AP EXAM CREDIT

BIOLOGY

You may also satisfy this requirement with a score/grade of:
4 or higher on the AP Biology exam satisfies BIOLOGY 1A & 1AL.

MATHEMATICS

You may also satisfy this requirement with a score/grade of:
3 or higher on the AP Calculus AB exam satisfies MATH 51;
3 or 4 on the AP Calculus BC exam satisfies MATH 51;
5 on the AP Calculus BC exam satisfies MATH 51 + 52.

READING AND COMPOSITION REQUIREMENT

You may also satisfy this requirement with a score/grade of:
4 or 5 on the AP exam in English Language and Composition satisfies READING & COMPOSITION A;
4 or 5 on the AP exam in English Literature and Composition satisfies READING & COMPOSITION A.

See ADDITIONAL REQUIREMENTS section for additional options to satisfy requirements.

LOWER DIVISION REQUIREMENTS

1 Complete A, B, C, D, E, F, G, and H

GENERAL CHEMISTRY WITH LABORATORY

A

CHEM 1A

General Chemistry

3.00

AND

CHEM 1AL

General Chemistry Laboratory

2.00

AND

CHEM 1B

General Chemistry

4.00

CHEM 1A

General Chemistry I

5.00

AND

CHEM 1B

General Chemistry II

5.00

AND

CHEM 1C

General Chemistry III

5.00

Regular and honors courses may be combined to complete this series

OR

CHEM 1AH

General Chemistry I - HONORS

5.00

AND

CHEM 1BH

General Chemistry II - HONORS

5.00

CHEM 1CH

General Chemistry III - HONORS

5.00

L

Regular and honors courses may be combined to complete this series

ORGANIC CHEMISTRY WITH LABORATORY

B

CHEM 3A Chemical Structure and Reactivity 3.00 AND CHEM 3AL Organic Chemistry Laboratory 2.00	← AND CHEM 12A Organic Chemistry I 5.00 CHEM 12B Organic Chemistry II 5.00
CHEM 3B Chemical Structure and Reactivity 3.00 AND CHEM 3BL Organic Chemistry Laboratory 2.00	← AND CHEM 12B Organic Chemistry II 5.00 CHEM 12C Organic Chemistry III 5.00

BIOLOGY

C

BIOLOGY 1A General Biology Lecture (Cells, Genetics, Animal Form & Function) 3.00 AND BIOLOGY 1AL General Biology Laboratory 2.00 L Lab is not required	BIOL 6A Form and Function in the Biological World 6.00 AND BIOL 6B Cell and Molecular Biology 6.00 ← OR BIOL 6AH Form and Function in the Biological World - HONORS 6.00 AND BIOL 6B Cell and Molecular Biology 6.00
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COMPUTER PROGRAMMING

Complete 1 course from the following.

D

ENGIN 7 Introduction to Computer Programming for Scientists and Engineers (MATLAB) 4.00	← No Course Articulated
COMPSCI 61A The Structure and Interpretation of Computer Programs 4.00	← No Course Articulated

MATHEMATICS

E

MATH 51 Calculus I 4.00	MATH 1A Calculus I 5.00 AND MATH 1B Calculus II 5.00 L Regular and honors courses may be combined to complete this series ← OR MATH 1AH Calculus I - HONORS 5.00 AND MATH 1BH Calculus II - HONORS 5.00 L Regular and honors courses may be combined to complete this series
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MATH 2B

Linear Algebra

5.00

L

Regular and honors courses may be combined to complete this series

←

OR

MATH 2AH

Differential Equations - HONORS

5.00

AND

MATH 2BH

Linear Algebra - HONORS

5.00

L

Regular and honors courses may be combined to complete this series

PHYSICS

F

PHYSICS 7A Physics for Scientists and Engineers 4.00	←	PHYS 4A Physics for Scientists and Engineers: Mechanics 6.00
PHYSICS 7B Physics for Scientists and Engineers 4.00	←	PHYS 4B Physics for Scientists and Engineers: Electricity and Magnetism 6.00 AND PHYS 4C Physics for Scientists and Engineers: Fluids, Waves, Optics and Thermodynamics 6.00
PHYSICS 7C Physics for Scientists and Engineers 4.00	←	PHYS 4C Physics for Scientists and Engineers: Fluids, Waves, Optics and Thermodynamics 6.00 AND PHYS 4D Physics for Scientists and Engineers: Modern Physics 6.00

READING AND COMPOSITION (R&C)

G

Courses that satisfy Reading & Composition A	←	ENGL C1000 Academic Reading and Writing 5.00 OR ENGL C1000H Academic Reading and Writing - HONORS 5.00 OR ESL 5 Advanced Composition and Reading 5.00
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ELECTIVES

H

END OF AGREEMENT